Dashcam License Plate Restoration Project Report

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Project Description:

My project's goal is to read the license plate numbers from images of license plates that have been corrupted by motion blurring either because they were taken from a moving camera such as a dashboard camera or because the car was moving very fast. My program does this in two parts. The first part is to perform image restoration on the motion blurred images. The program receives as input a picture that is already cropped closely to the license plate that is to be processed along with an estimation of the motion blur parameters. It then uses these parameters to construct an estimation of the degradation function h(x, y) and uses the DFT of the estimated degradation function H(u, v) to create a Wiener filter that would be used to compute the restored image as shown in the equation below.

$$\hat{f}(x,y) = \mathcal{F}^{-1}\left\{\frac{H^*(u,v)}{|H(u,v)|^2 + K}G(u,v)\right\}$$

with:

- $H(u,v) = \mathcal{F}\{h(x,y)\}$ where $h(x,y)$ is the estimated degradation function
- $G(u,v) = \mathcal{F}\{g(x,y)\}$ where $g(x,y)$ is the input image
- $K = {}^{1}/_{SNR}$ where SNR is the estimated signal to noise power ratio

The second part of my program uses a template matching technique to "read" the number on the license plate. It starts by using Otsu's method thresholding to convert the restored image to a binary black and white image. Then morphological erosion with a vertical filter is applied, along with morphological restoration by opening to remove smaller text that is not the license number. Next, column totals are used to identify "valleys" in the image where all values in the given columns are zero. These valleys represent the gaps between the characters, and they can be used to segment each character into a separate image. Finally, each character is compared to a template library of binary license plate character images of the same font in order to determine what the character is. Each comparison uses normalized cross-correlation to generate a match score, and whichever template image produces the highest max score will be taken as the character. It will create a string of the detected characters, and this will be the final output.



Figure 1. Project Dataflow Diagram

Constraints & Performance:

As mentioned previously, the program is designed for an input image which is already cropped closely to the license plate itself. For the sake of simplicity, I also chose to focus on one license plate format, the Texas general issue design which is currently the standard license plate format in the state of Texas and is pictured below. The program's performance has not been tested outside of these bounds.



Figure 2. Texas General Issue License Plate Format

The program preforms optimally when the length of the image blur is about 10% or less of the relevant axis of the image according to the blur angle. For example, the blur can have a higher pixel length if it is applied horizontally since the license plate is about twice as long as it is tall. Therefore, the optimal motion blur threshold is also highly dependent on the resolution of the image. Testing was done with a k value of 0.005 and using length and angle parameters of the images that were manually blurred, so discrepancies in the accuracy of these parameters would also cause a decrease in the accuracy of the results.

Usage:

All of my project files are included in the google drive folder which I have pasted a link to in the project submission as well as the end of this document. The folder's contents are as follows:

- Three MATLAB code files each including a separate function
    - apply_wiener_filter.m
        - Input: blurred image and blur parameters
        - Output: unblurred image achieved with a wiener filter
    - segment_plate_text.m
        - Input: unblurred image
        - Output: binary image with any text or noise smaller than the plate numbers removed using mathematical morphology techniques
    - recognize_license_plate.m
        - Input: binary image with isolated license number

        - Output: segmented characters with along with the license number string
- sample license plate images with and without motion blur applied
    - images without blur are labeled ‘sampleX’
    - images with blur are labeled ‘sampleX_blurred(YpZdegrees)’ where Y and Z represent the motion blur parameters, Y in pixels and Z in degrees
- Template images for each possible license number character (A-Z, 0-9)

When all of these files are contained in the same folder, the program may be executed in MATLAB by executing the following commands in the command window:

```
restored_image = apply_wiener_filter('blurred_image.png', L, theta, k);
filtered_image = segment_plate_text(restored_image);
recognized_plate = recognize_license_plate(filtered_image);
disp(['The recognized license plate number is: ', recognized_plate]);
```

For example:

```
>> restored_image = apply_wiener_filter('sample1_blurred(50p45degrees).jpg', 50, 45, 0.005);
filtered_image = segment_plate_text(restored_image);
recognized_plate = recognize_license_plate(filtered_image);
disp(['The recognized license plate number is: ', recognized_plate]);
```

Figure 3. Execution Command Example

Google Drive Folder:

https://drive.google.com/drive/folders/19cwX_lOjiuasrb0INRz7Cs5Zp3wX7ICq?usp=sharing

I have shared the folder with the professor, TA, and grader as well as included the link on the submission page, if there are any issues accessing it you can contact me at josiahtate@tamu.edu and I will resolve the issue as quickly as possible.